Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Aim: To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:

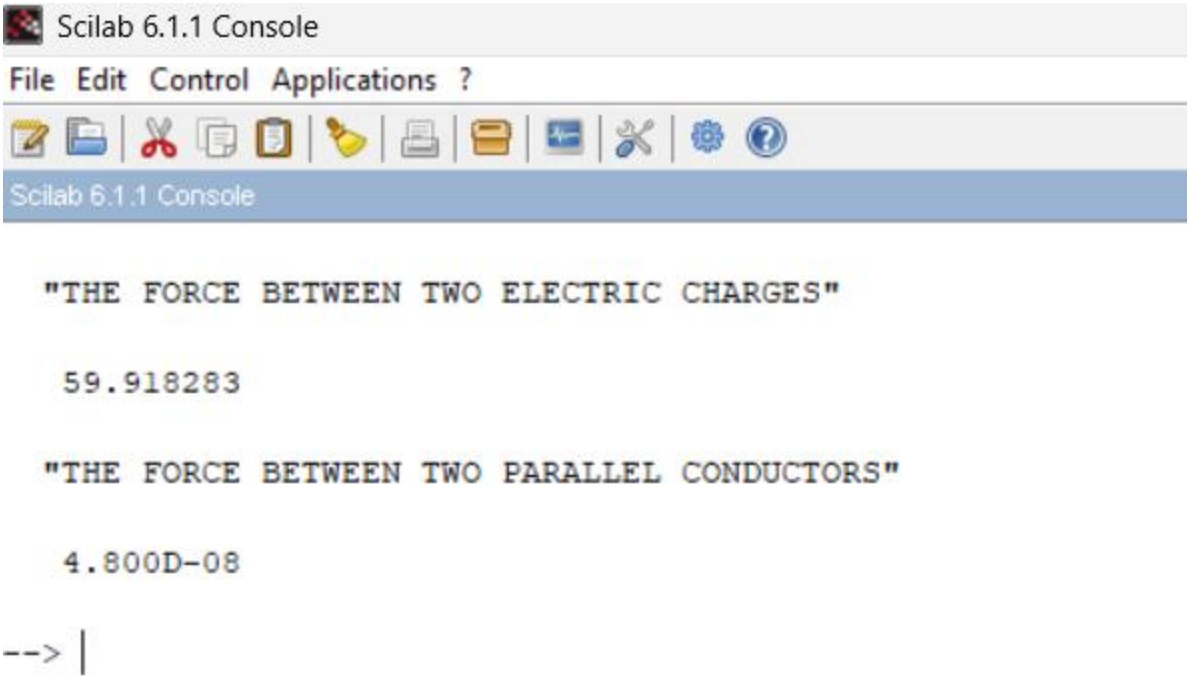
exp-3(M) (C:\Users\ramac\exp-3(M)) - SciNotes

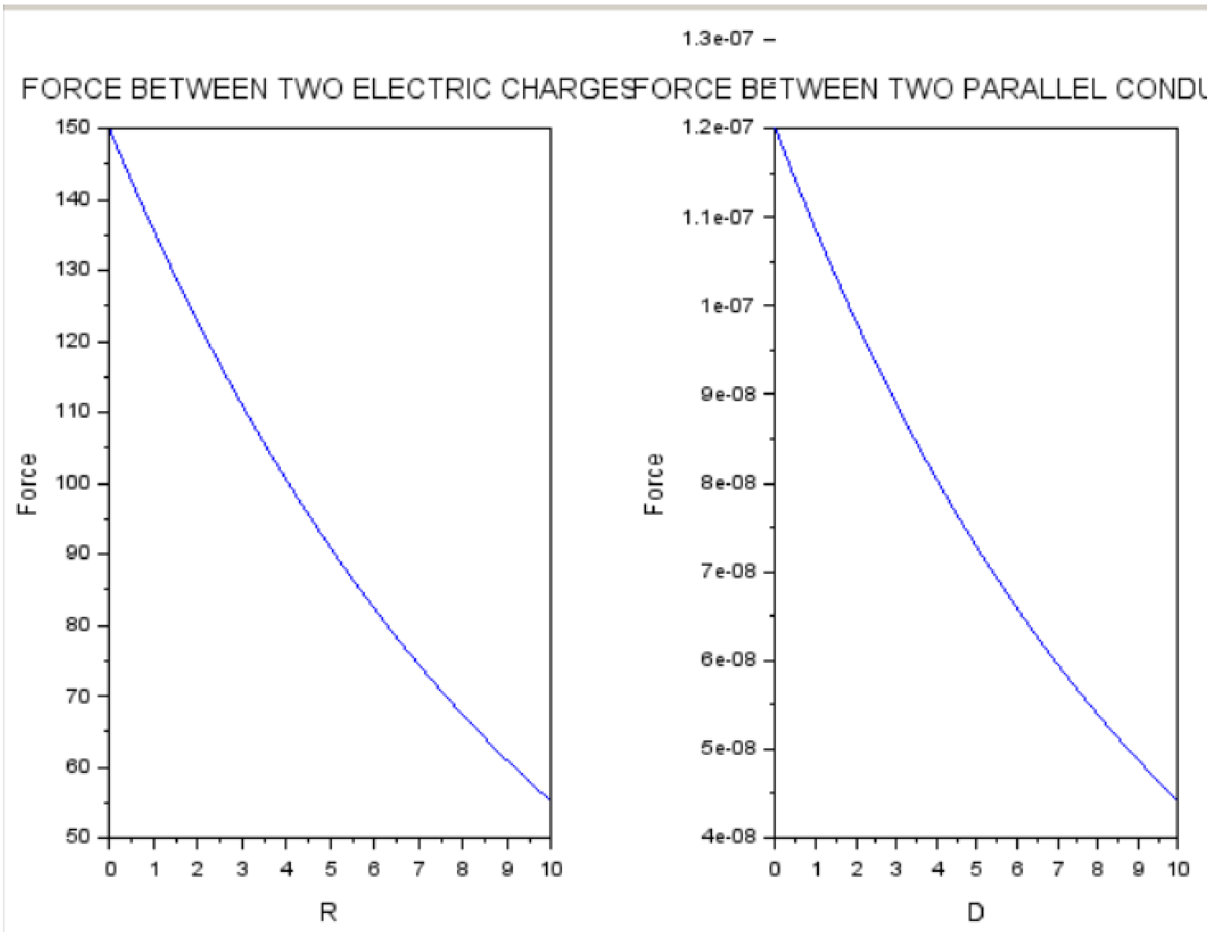
File Edit Format Options Window Execute ?

exp-3(M) (C:\Users\ramac\exp-3(M)) - SciNotes

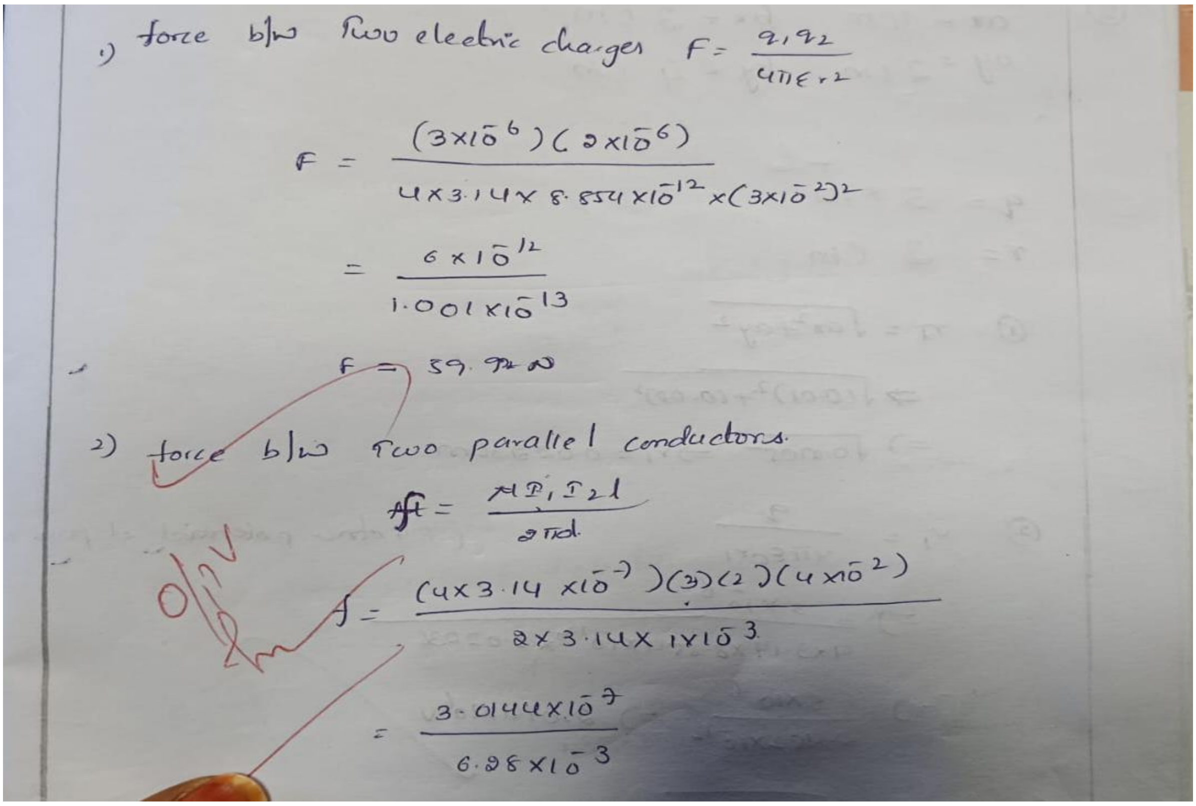
emf-1(2) | emf-1(3) | 2=Electric field and Electric flux density | 2=case(2) | 2-case(3) | 333333 | 2-case(1)

```
clc;
clear;
clf();
charge_1 = 3*10^-6;
charge_2 = 2*10^-6;
radius = 3*10^-2;
current_1 = 3;
current_2 = 2;
l = 4*10^-5;
d = 1*10^-3;
mu = 4*%pi*10^-7;
epsilon = 8.854*10^-12;
F = (charge_1*charge_2)/(4*%pi*epsilon*radius^2);
f = (mu*current_1*current_2*l)/(2*%pi*d);
disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES");
disp(F);
disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS");
disp(f);
x = linspace(0,10,50);
y = F*exp(-0.1*x);
z = f*exp(-0.1*x);
subplot(1,2,1);
plot(x,y);
xlabel("R");
ylabel("Force");
title("FORCE BETWEEN TWO ELECTRIC CHARGES");
subplot(1,2,2);
plot(x,z);
```


Scilab 6.1.1 Console
File Edit Control Applications ?
Scilab 6.1.1 Console
"THE FORCE BETWEEN TWO ELECTRIC CHARGES"
59.918283
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"
4.800D-08
-->


FORCE BETWEEN TWO ELECTRIC CHARGES
Force
R
150
140
130
120
110
100
90
80
70
60
50
0 1 2 3 4 5 6 7 8 9 10
FORCE BETWEEN TWO PARALLEL CONDU
Force
D
1.3e-07
1.2e-07
1.1e-07
1e-07
9e-08
8e-08
7e-08
6e-08
5e-08
4e-08
0 1 2 3 4 5 6 7 8 9 10

Theoritical Output:



1) force b/w two electric charges $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$

$$F = \frac{(3\times10^{-6})(2\times10^{-6})}{4\times3.14\times8.854\times10^{-12}\times(3\times10^{-2})^2}$$

$$= \frac{6\times10^{-12}}{1.001\times10^{-13}}$$

$$F = 59.92\ N$$

2) force b/w two parallel conductors.

$$F = \frac{\mu I_1 I_2 l}{2\pi d}$$

$$f = \frac{(4\times3.14\times10^{-7})(3)(2)(4\times10^{-2})}{2\times3.14\times1\times10^{-3}}$$

$$= \frac{3.0144\times10^{-7}}{6.28\times10^{-3}}$$

Result : Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.